Arush Manem

EDUCATION

University of Minnesota, Twin Cities — ***Minneapolis, MN***

Bachelor of Science, Honors Program — Data Science and Computer Science (GPA: 3.8) — *May 2027*

SKILLS

Technical: Python, OCaml, SQL, PyTorch, Java, C/C++, R, FastAPI, React, Node.js, TypeScript, HTML/CSS, NumPy, Pandas, PySpark, Matplotlib, ETL Pipelines, Machine Learning, REST APIs, Docker, GitHub, Jira, Power BI, PowerShell, DevOps, RStudio, LangChain, ChromaDB, OpenAI API, RAG, Embeddings

EXPERIENCE

UnitedHealth Group (UHG) — ***Eden Prairie, MN***

Consumer Resolution Center War Room Intern — *June 2026 - August 2026*

- Engineered agentic automation pipelines in Python and SQL on Databricks to scale intelligent triage capabilities for the CRC War Room, a Harvard Business Review-featured enterprise healthcare team
- Contributed automation and data infrastructure whose outputs directly enabled the War Room's 2026 outcomes: 6.2M members impacted and 1.4M calls obviated enterprise-wide, with agentic triage expansion being a driving force of operational scale
- Built real-time dashboards and operational data widgets in Power BI and ServiceNow to surface resolution outcomes and member impact signals across high-volume healthcare workflows
- Partnered with Project Managers and Triage Analysts to translate operational bottlenecks into technical requirements, delivering end-to-end automation and data solutions across enterprise problem-solving workflows

HealthPartners — ***Bloomington, MN***

Software Engineering Intern — Robotic Process Automation — *June 2022 - January 2026*

- Engineered scalable automation systems in UiPath/VB.NET to streamline claims processing, QA, and multi-system testing, reducing execution cycles and saving 37,000+ hours per year collectively
- Collaborated with developers, business analysts, and stakeholders as part of a team collectively saving $2M+ and producing $3.5M+ in additional revenue; contributed to a 3-year projected ROI of $3.86M
- Built and maintained ETL pipelines to extract, transform, and analyze healthcare claims data (4,000+ records run) for operational analysis, and contributed to automating 1.3M+ tasks previously handled by humans
- Automated web testing workflows for UI regression coverage, shortening release cycles and improving deployment efficiency by 90%
- Explored AI-assisted orchestration (UiPath Maestro), integrating intelligent decisioning into multi-system processes — early experience with intelligent, distributed automation

PROJECTS

SnapCount | Python, React, FastAPI, PyTorch, Pandas, Recharts, Google Gemini API — ***December 2025***

NFL Fantasy Analytics Platform

- Engineered a full-stack predictive analytics platform using FastAPI and React, integrating robust data processing with a responsive user interface
- Developed an automated ETL pipeline to scrape, clean, and cache 5,000+ weekly records using nfl_data_py and Pandas, ensuring real-time data availability
- Designed a custom statistical algorithm to quantify defensive metrics and forecast player performance, processing historical variance to generate Start/Sit recommendations
- Built a deep learning inference pipeline using PyTorch, replacing legacy rule-based logic with a neural network to forecast player performance based on historical matchups and home-field advantage
- Integrated Google Gemini LLM via Python to generate context-aware, natural language player recommendations, with robust error handling and retry logic for API rate limits

Insurance Risk Assessor | Python, Jupyter, scikit-learn, SHAP — ***July 2025***

Machine Learning System

- Built a modular ETL pipeline for preprocessing, outlier detection, and feature generation (smoker–age and BMI–age interactions); trained and optimized a Random Forest regression model achieving $R^2 = 0.88$ and MAE ≈ $2,600
- Packaged the model with Pickle for reproducible, real-time deployment, and applied SHAP explainability to interpret feature impacts for stakeholders

AI Operations Assistant | Python, LangChain, ChromaDB, OpenAI API — ***Summer 2026***

Retrieval-Augmented Generation (RAG) System

- Built a modular RAG pipeline from scratch with PDF ingestion, character-level chunking with overlap, embedding generation (text-embedding-3-small), ChromaDB vector storage, and gpt-4o-mini answer synthesis organized across a clean multi-layer architecture
- Engineered a retrieval evaluation harness across 20 adversarial and factual test cases, measuring Recall@k, MRR (0.873), LLM-as-judge scores, and adversarial refusal rate (100%)
- Improved answer quality through iterative prompt engineering, raising LLM-as-judge scores from 4.35 → 4.65/5 and resolving targeted failure cases in multi-entity synthesis queries

LEADERSHIP & PROFESSIONAL DEVELOPMENT

Data Science Club — **Minneapolis, MN**

Active Member — *September 2024 – Current*

- Presented insights on cutting-edge data analytics methodologies in industry forums
- Engaging in industry networking and professional development through meetings with insurance leaders, career fairs, and workshops